

01:XXX:XXX - Homework n

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1. Periodic 1D model. Continuous space and Discrete Time. Take one "band" on space condition on that and diffuse out to next to get the next one.
2. $\sum A_j(t) \cos + B_j(t) \sin$ Allow $\text{Cov } A_{j,t}, A_{j,t+k} = \sigma_j^2 \rho_j^k$ And the faster σ_j^2 decay in space control the smoothness
 Construct the ρ_j with exponential and then see from there
 Overhethinger - fourier series thin red book
3. Seperability is not the issue, but it is the smoothness away from the origin.
4. Screening effect is also a symptom of non smoth covariance
5. Use Power Series (maybe alot of terms) to see if there is weirdness in short lags.
6. Random fourier Series, and can work out covariance function explicitly.
7. Given conditional, and prior step, is this spacetime process essentialy markov.